



Faculty of Engineering and Computer Science
Department of Electrical and Computer Engineering

MISSION STATEMENT

ELEC/COEN 390 - BB

Electrical/Computer Engineering Product Design Project

Stakeholders

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**We certify that this submission is our original work and meets the
Faculty's Expectations of Originality**

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Automated Aquarium Monitoring Systems (AAMS Inc.)

AQUASENSE

Mission Statement • Rev. 1.0.0

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1. Product Description

AquaSense is a real-time aquarium monitoring system that helps owners maintain safer environments for their marine animals. It integrates four sensors: pH, temperature, TDS (total dissolved solids), and water level into a packaged, attachable sensor unit that connects to an easy-to-use Android app. Readings are captured by an Arduino Uno R4 with built-in WiFi, pushed to a Firebase cloud database, and displayed live in the app, which fires threshold alerts when conditions become critical. One app manages multiple aquariums with instant alerts and historical analytics.

2. Benefit Proposition

By automating water quality monitoring, AquaSense frees users from difficult, time-consuming manual testing and prevents tools from being disconnected without the user's knowledge. Real-time readings, instant critical-condition alerts, and easy-to-read analytics let aquarium owners stay on top of every tank and act before small changes threaten marine life. From data collection to alerts and graphical analysis, all in one place.

3. Key Business Goals

Our main business goal was to develop a working, stable prototype that combines real-time sensor data with an intuitive Android app. We focused on four objectives: protect (alert users about the safety of their pets), simplify (make water parameters easy to understand), connect (link the hardware to a cloud database and mobile app), and analyse (store and graph data to guide further aquarium improvements). While AI-driven features such as predictive spoilage or automated recommendations were considered as future enhancements, the current prototype specializes in the baseline tools that directly answer user needs.

Over the course of the project, we budgeted a total of 300 ideal hours across all three sprints. Based on a development cost of \$40/hour, our team investment represents approximately \$12,000 of development effort.

On the hardware side, sensor components cost approximately \$40 per unit, with a suggested sale price of \$50, yielding a gross margin of \$10 per unit. Break-even would occur at approximately 240 units sold. Given the growing aquarium hobbyist market and the long-term cost of maintaining aquariums and animal care, we consider this a reasonable and realistic goal.

4. Target Market

AquaSense is intended for two primary groups. The first is hobbyists — casual aquarium owners looking to take the next step in improving their aquarium maintenance. The second is business owners who balance multiple aquariums and need a better automated system to track water quality across all of them. Both groups value automated monitoring, instant alerts, and instant feedback on tank conditions without the burden of constant manual testing.

5. Assumptions

We made the following assumptions at the start of the project:

- **Users would trust automated sensor readings over manual test kits.** Stakeholder interviews confirmed that continuous, hands-off monitoring and live updates were seen as a major advantage, with one aquarium owner noting the live updates made it very easy to stay on top of each of their tanks.
 - True, automated sensors were preferable.
- **The four sensors would reliably detect out-of-range conditions across different tank types.** The combination of pH, temperature, TDS, and water level probes provided valuable real-time monitoring and triggered useful threshold alerts.
 - True, the sensors were reliable in sending data

As the project evolved, additional assumptions emerged and were tested:

- **A single universal sensor set would suit all aquarium types.** Partially true — research showed that saltwater, freshwater, and other tanks require different ideal ranges for pH, TDS, and temperature. We addressed this with aquarium templates that set appropriate thresholds per tank type, though a wider range of species profiles could improve accuracy.
 - False, an additional setting where users could customize threshold values was added to address all possible tank types

6. Constraints

Several constraints shaped the development of AquaSense. The application was built for Android only, as cross-platform development was not feasible within the time and cost limits. AI-driven features such as predictive water quality forecasting and automated recommendations were left out of scope due to development time and complexity. The system currently focuses on core monitoring, with room to expand species profiles and account features. The app depends on a persistent internet connection to communicate with Firebase, limiting offline capability. Finally, the sensor hardware requires a power source and periodic maintenance; integration into a permanent, power-connected enclosure remains an area for future improvement.

7. Stakeholders

The most important stakeholders for AquaSense are aquarium owners — both hobbyists and business owners managing multiple tanks. These users benefit directly from live monitoring, threshold alerts, and historical analytics that help them protect their marine animals and maintain stable conditions. We conducted 15 rounds of interviews with aquarium owners, receiving overall positive feedback, with the analytics feature praised in particular for its visual data representation. Aquarium retailers and equipment manufacturers are future stake-

holders who could collaborate on sensor bundling, product recommendations, or integrated hardware, extending the system's value and reach.